

1046Hz - SYNTHANDS

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COMPUTER MUSIC: LANGUAGES AND SYSTEMS - FINAL PROJECT

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Outline

- 1 Overview
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- 3 Computer music unit
- 4 Graphical feedback unit
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Overview

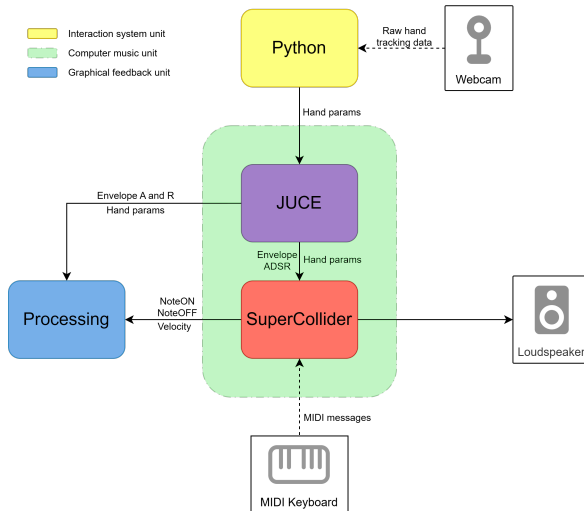
Project Goal

SYNTHANDS

Gesture driven digital musical instrument that uses webcam-based hand tracking to shape the sound.

- Python: real-time gesture analysis and data processing
- JUCE: interface enabling user control and forwarding parameters
- SuperCollider: main sound synthesis and audio processing engine
- Processing: dynamic visual representation

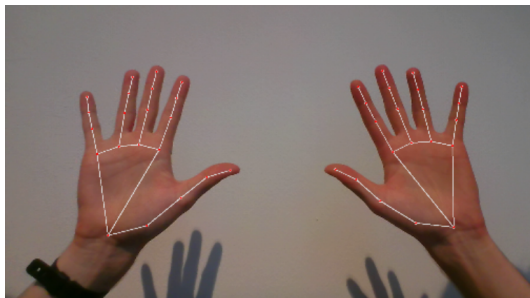
Data Flow



Interaction system unit

MediaPipe

The MediaPipe framework was used to implement the hand tracking module, which provides real-time hand landmark detection and tracking capabilities.



Real time hand tracking anchor points visualization

Python modules

- main.py
- config.py
- hand_tracker.py
- gesture_features.py
- smoothing.py
- osc_sender.py

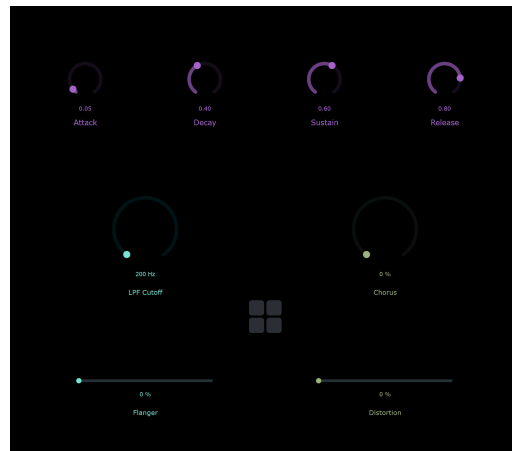
```
[OSC] /hand/left/open -> 0.104  
[OSC] /hand/left/open -> 0.103  
[OSC] /hand/left/open -> 0.103  
[OSC] /hand/left/open -> 0.106  
[OSC] /hand/left/open -> 0.101  
[OSC] /hand/left/open -> 0.127  
[OSC] /hand/left/open -> 0.325  
[OSC] /hand/left/open -> 0.527  
[OSC] /hand/left/open -> 0.669  
[OSC] /hand/left/open -> 0.768  
[OSC] /hand/left/open -> 0.838  
[OSC] /hand/left/open -> 0.887  
[OSC] /hand/left/open -> 0.921  
[OSC] /hand/left/open -> 0.944  
[OSC] /hand/left/open -> 0.961  
[OSC] /hand/left/open -> 0.973  
[OSC] /hand/left/open -> 0.981
```

Debug mode to visualize the OSC messages sent to the
JUICE GUI

Computer music unit

JUCE GUI

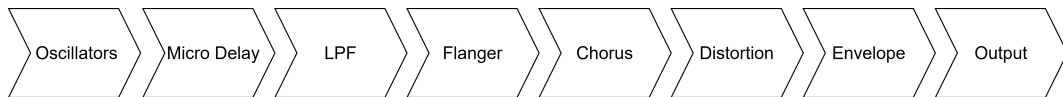
- LPF Cutoff: left hand openness
- Flanger: left thumb position
- Chorus: right hand openness
- Distortion: right thumb position
- ADSR knobs: only manual control



Default GUI values when opened

Supercollider sound engine

SuperCollider is the audio engine of the project. It receives control data from JUCE interface and combines predefined SynthDef parameters with dynamic control routines. Filtering techniques to smooth and stabilize the signal are involved.



Supercollider signal flow scheme

Graphical feedback unit

Processing

A Processing-based interface provides real-time visualization of hand gestures and audio control parameters, in particular:

- Hand and thumb related parameters from Python through JUCE
- Attack and release parameters from manually set on the ADSR envelope in JUCE
- Velocity, NoteON and NoteOFF from MIDI through supercollider

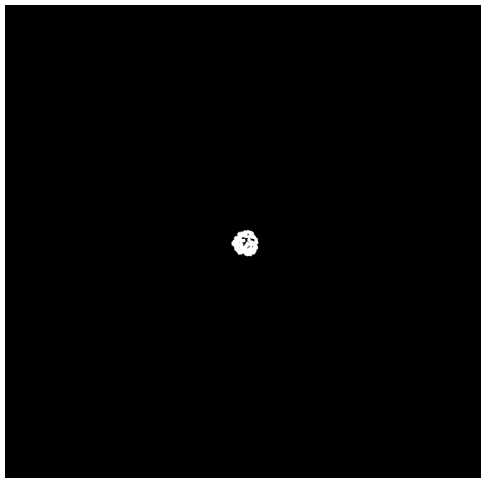


Figure 1: NoteOFF visualization

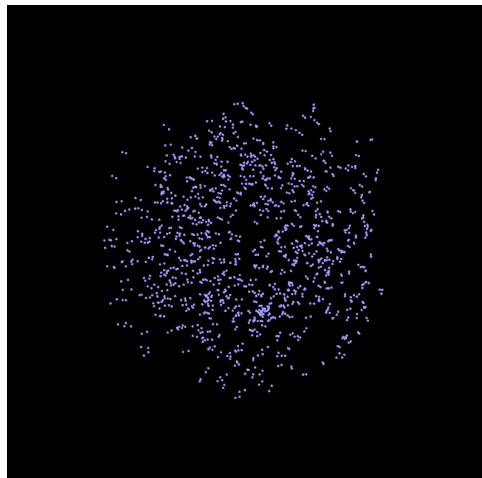


Figure 2: NoteON visualization

Live demonstration